

Week 12 - Milestone Two week

1. Lesson: no lesson this week. Milestone Two is due!

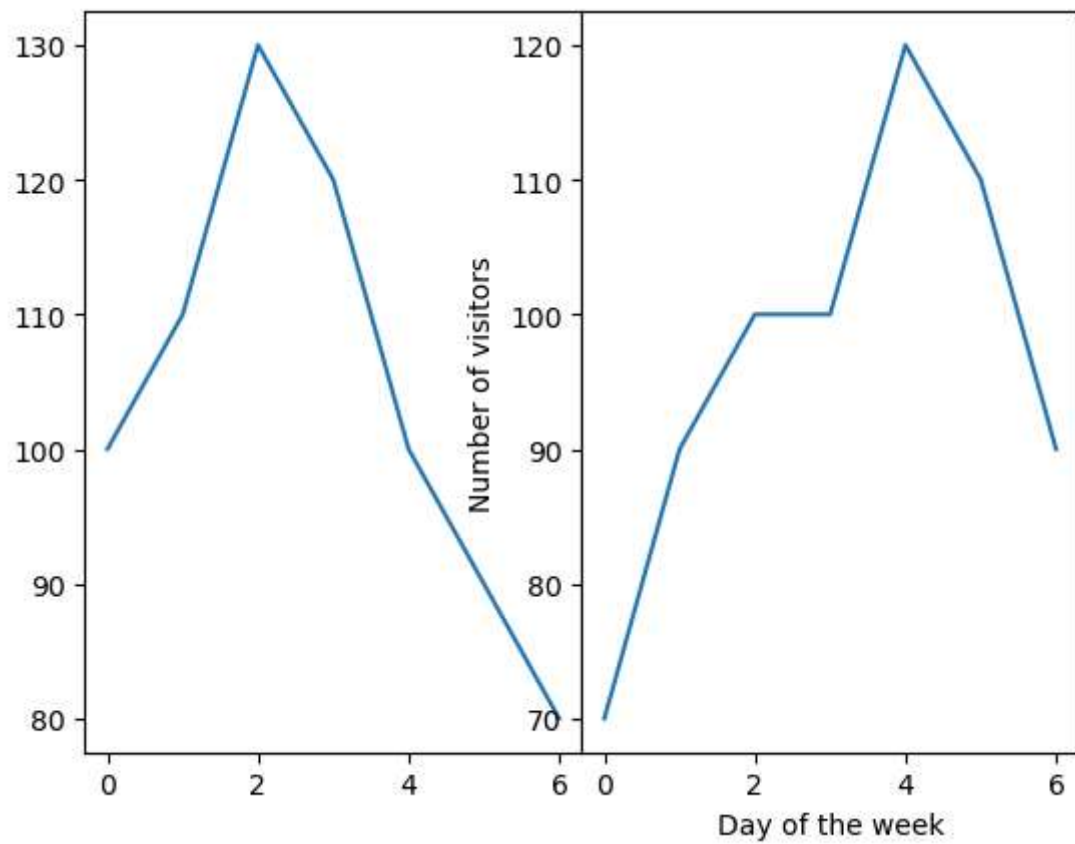
2. Weekly graph question

Suppose you plot the number visitors to a museum over the seven days of a week; then again the following week. Can you suggest the advantages and disadvantages of the following two ways of presenting the data?

```
In [1]: import numpy as np
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.ticker import FuncFormatter
```

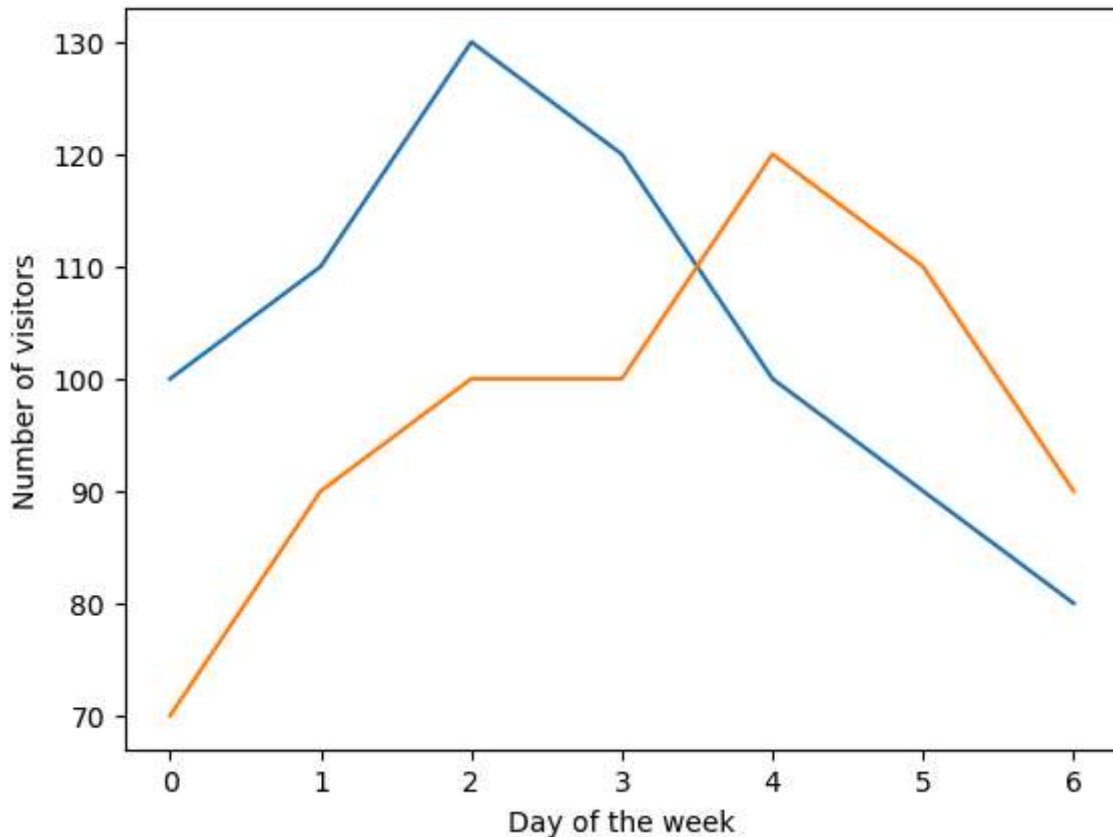
```
In [2]: np.random.seed(0)
time_series_0 = np.array([100, 110, 130, 120, 100, 90, 80])
time_series_1 = np.array([70, 90, 100, 100, 120, 110, 90])
fig, axs = plt.subplots(1, 2)
axs[0].plot(time_series_0)
axs[1].plot(time_series_1)
plt.subplots_adjust(wspace = 0)
plt.xlabel("Day of the week")
plt.ylabel("Number of visitors")
```

```
Out[2]: Text(0, 0.5, 'Number of visitors')
```



```
In [3]: plt.plot(time_series_0)
plt.plot(time_series_1)
plt.xlabel("Day of the week")
plt.ylabel("Number of visitors")
```

```
Out[3]: Text(0, 0.5, 'Number of visitors')
```



3. Working on your datasets

This week, you can work on Milestone Two rather than having a particular exercise about working on your datasets

4. Storytelling With Data plot

Reproduce any graph of your choice in p. 187-205 of the Storytelling With Data book as best you can. (Chapter Eight). You do not have to get the exact data values right, just the overall look and feel.

```
In [4]: data = pd.DataFrame({
    'Year': [2019, 2020, 2021, 2022, 2023, 2024, 2025,
             2019, 2020, 2021, 2022, 2023, 2024, 2025,
             2021, 2022, 2023, 2024, 2025,
             2022, 2023, 2024, 2025,
             2024, 2025],
    'Product Type': ['Product A', 'Product A', 'Product A', 'Product A', 'Product A',
                    'Product B', 'Product B', 'Product B', 'Product B', 'Product B',
                    'Product C', 'Product C', 'Product C', 'Product C', 'Product C',
                    'Product D', 'Product D', 'Product D', 'Product D',
                    'Product E', 'Product E'],
    'Value': [390, 420, 425, 385, 300, 285, 280,
             370, 400, 410, 380, 290, 275, 270,
```

```

        100, 190, 200, 225, 190,
        150, 260, 215, 210,
        100, 205]
    })

```

```

In [5]: fig, ax = plt.subplots()

sns.barplot(data = data, x = 'Product Type', y = 'Value', hue = 'Year', palette = '

ax.legend(loc = 'lower center', bbox_to_anchor = (0.5, -0.18), ncol = 7, frameon =

ax.set_title('Average Retail Product Price Per Year', fontweight = 'bold')

ax.set_yticks(list(range(0, 600, 100)))
ax.yaxis.set_major_formatter(FuncFormatter(lambda y, _: f"${y:,.0f}"))
ax.set_ylabel('')
ax.set_xlabel('')

plt.show()

```

